



Unraveling the molecular significance of CYP1A2 (rs762551; c.-9–154 C>A) genetic variant on breast carcinoma susceptibility: Insights from case-control study and meta-analysis

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ARTICLE INFO

Keywords:

Cytochrome P450
CYP1A2*rs762551
Genetic variants
Breast carcinoma

ABSTRACT

Background: The human cytochrome P450 (CYP) superfamily encompasses different categories of isoenzymes that contribute to multiple metabolic processes involving drug detoxification, cellular signaling, and the proliferation of malignant tissues. Using genetic technology, customized bioinformatic analysis, and meta-analysis design, the main goal of this study was to identify the association between the CYP1A2*rs762551 variant and the susceptibility to breast carcinoma (BRCA).

Methods: The case-control study was conducted based on 104 BRCA women and 102 healthy controls. Using the TaqMan allelic discrimination assay, the CYP1A2 (rs762551; c.-9–154 C>A) variant was genotyped. Bioinformatic frameworks and logistic regression analysis were used to assess the involvement of this genetic variant in BRCA development. A meta-analysis design was accomplished based on our case-control study and other previously published records. Publication bias, heterogeneity between studies, and trial sequential analysis (TSA) were analyzed.

Results: The CYP1A2*rs762551 variant conferred protection against BRCA development under allelic (OR = 0.48, *p*-value < 0.001), dominant (OR = 0.34, *p*-value < 0.001), and recessive (OR = 0.44, *p*-value = 0.011) models. However, this intronic variant was correlated with a decreased risk of BRCA among late-onset menopause women compared to other cases. Bioinformatic analysis confirmed that this genetic variant has a functional impact on the progression of tumorigenesis. Moreover, this meta-analysis design included 12922 BRCA women and 15603 healthy controls. Our findings disclosed the contribution of the CYP1A2*rs762551 variant with protection against cancer development among Caucasian females under allelic (OR = 0.75, *p*-value = 0.025), and dominant (OR = 0.58, *p*-value = 0.015) models.

Conclusions: This case-control study confirmed the contribution of the CYP1A2*rs762551 variant with decreased risk of BRCA development among Egyptian subjects. Moreover, BRCA women with late-onset menopause conferred protection against cancer progression compared to other subjects. Our findings identified that this meta-analysis design achieved protection against BRCA development among Caucasian women compared to other ethnicities.

1. Introduction

Breast carcinoma (BRCA; MCID#: BRS047) is referring to mammary

carcinoma, that belongs to one of the most adverse malignant tumors within adult females, in which the mammary tissues begin to extend and grow in an uncontrolled and peculiar fashion [1,2]. Based on the

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<https://doi.org/10.1016/j.prp.2024.155501>

Received 20 May 2024; Received in revised form 13 June 2024; Accepted 27 July 2024

Available online 5 August 2024

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enlisted statistics provided by the World Health Organization (WHO), there were more than two million women diagnosed with BRCA and about 700,000 fatalities within the whole population [3,4]. Additionally, the International Agency for Research on Cancer (IARC) pronounced that the absolute incidence numbers among Egyptian subjects were 26,845 (34.9 %) BRCA females compared to other cancer types [5]. Other studies conducted among Egyptian women found that 46,000 BRCA women were expected to be affected by this malignant disease in the next two decades [6,7]. The estimated prevalence rate of BRCA cases among Egyptian women was 19,450 (37.7 %), which was underestimated compared to other populations due to numerous restrictions pertaining to monitoring and registration capabilities [8,9]. Moreover, the absolute numbers of fatalities among Egyptian women were 9596 (22.0 %) compared to other types of carcinomas [6]. Thus, a comprehensive understanding of the pathophysiology and fundamental mechanisms concerning breast malignancies is crucial to uncovering the applicable methods used in the diagnostic and prognostic techniques [10,11]. Additionally, various reports were conducted to study the genetic profiling, epigenetic factors, and hereditary variants associated with the progression of breast carcinoma, but with challenging conclusions [12,13].

The human cytochrome P450 (CYP) superfamily comprises different enzymatic isoforms that could oxidize enormous compounds of xenobiotic and endobiotic elements [14]. These isoenzymes might execute potential operations in metabolizing xenobiotic drugs, synthesis of various steroid hormones, caffeine detoxification, activation of polyunsaturated fatty acids, and accelerating the probable mechanism of fat-soluble vitamins [15,16]. Moreover, these CYP enzymes have causative responsibilities in the transformation of polycyclic aromatic hydrocarbons and reactive metabolites into carcinogenic chemicals that enhance the pathogenesis of various tumorigenesis, particularly breast carcinoma [15,17]. Interestingly, the cytochrome P450 family 1 subfamily A member 2 (*CYP1A2*; ENSG00000140505) gene encodes a specific enzyme that exerts a vital function in the metabolic processes related to numerous lipophilic compounds and xenobiotic metabolites [18,19]. The *CYP1A2* enzyme is classified as a monooxygenase subfamily that might incorporate a single atom of molecular oxygen inside their metabolites, leading to the hydroxylation of specific drugs and the metabolization of various environmental procarcinogens that modify genomic DNA and produce certain DNA adducts and mutagenic carcinogens [20,21]. Furthermore, the *CYP1A2* gene is positioned on the elongated arm of chromosome number #15q24.1 and is made up of 7763 bases that are arranged and transversed along the plus strand [13, 20]. Interestingly, numerous reports have explored the contribution of the *CYP1A2* (*rs762551*; c.-9-154 C>A) variant to the elevated risk of various tumors, involving prostate cancer [22], gastric cancer [23], lung carcinoma [24], colorectal cancer [25], bladder carcinoma [20], and breast carcinoma [10,26–28]. This polymorphic variant is situated in the promoter region at the first intron of the *CYP1A2* gene, and thought to be correlated with modulation of enzymatic inducibility, in which the C allele was associated with low enzymatic activity, while the A allele connected with higher activity of the enzyme [27,29]. Consequently, this research was conducted to elucidate the involvement of the *CYP1A2* (*rs762551*; c.-9-154 C>A) variant with the susceptibility to breast carcinoma among Egyptian females using genetic analysis, tailored bioinformatic methods, and further meta-analysis design.

2. Subjects and methods

2.1. Case-control study

2.1.1. Study participants and ethical agreement

This hospital-based work was carried out at the Outpatient's Clinic of the Medical Oncology Center, Mansoura University, Egypt. The study design is comprised of 206 participants [104 BRCA women and 102 unrelated healthy controls] recruiting from the same geographic

territory and matched with age and menopause status. The gold-standard diagnostic procedure for breast carcinoma was histologically operated on by specialized oncologists using a hollow tube of fine needle biopsy, as described previously [8]. BRCA women were categorized into two distinct groups, including 37 (35.6 %) women with premature menopause and 67 (64.4 %) women with late-onset menopause. Briefly, premature menopause is a term that defines females who started to menopause prior to the age of 45 years, while late-onset menopause stands for females who didn't establish menopause at 55 years or older [30]. Additionally, BRCA women with a history of tumorigenesis, autoimmune illnesses, chemotherapy administration, renal difficulties, diabetic issues, and mammary breastfeeding were ruled out from this study. The staging protocols for breast carcinoma were processed based on the American Joint Committee on Cancer (AJCC) for assessment of the TNM staging system [31]. This study obtained an ethical agreement from the scientific committee of Institutional Review Board (IRB) for human investigation. All the participants enrolled in this work were encouraged to provide a written informed consent after acquiring the accredited information about the advantages of hereditary screening in predicting breast carcinoma. Furthermore, this research was conducted according to Helsinki's recommendations and their guidelines.

2.1.2. Selection criteria for the *CYP1A2***rs762551* variant

Briefly, our team was motivated to study the pathogenic impact and functional prognosis for cytochrome P450 genes with the susceptibility to breast carcinoma; we performed a systematic screening and bioinformatic investigation to select the common SNP polymorphic within the *CYP1A2* gene. Based on the Ensembl Genomic Database (<https://www.ensembl.org/>), the most common allelic variant identified in the *CYP1A2* gene with MAF ≥ 0.4 was the *rs762551* (*ENST00000343932*; c.-9-154 C>A) variant. Additionally, we explored multiple electronic repositories to retrieve important data concerning the functional analysis of this potential gene and its corresponding protein including the human gene-centric database; GeneCards (www.genecards.org), the National Center for Biotechnology Information; NCBI-Gene database (<https://www.ncbi.nlm.nih.gov/gene/>), String database for visualization the interactions between different target proteins using genomic ontology (<https://string-db.org/>), the Protein Data Bank for tertiary configuration of biomolecules (<https://www.rcsb.org/>), Compartments database for protein subcellular localization evidence (<https://compartments.jensenlab.org>), GeneMania platform for constructing hypotheses regarding genomic function (<https://genemania.org/>), and MEXPRESS database for precise integration of copy number variations based on TCGA data portal (<https://mexpress.be>) [32]. Lastly, the potential regulatory annotations of this non-coding variant were determined using chromatin interactions in the tertiary genomic structure with multiple genetic variants. This retrieved information was extracted from the 3DSNP v2.1 database, which linked the non-coding variants to tertiary visualization of interacting genes (<https://3dsnp.omic.tech/>) [33]. All these electronic repositories were accessed on February 26, 2024.

2.1.3. Sample collection and DNA extraction

The manipulation and collection of peripheral whole blood samples from the enrolled participants were conducted under sterilized conditions. These collected samples were subjected to being combined with EDTA-anticoagulant that was designed for hereditary analysis. Additionally, the genomic DNA was managed and extracted based on the manufacturer instructions provided by the GeneJET Mini-Kit for whole blood DNA purification (Thermo Scientific™, Waltham, MA, USA). Moreover, the purification of genomic DNA was measured utilizing the NanoDrop® ND-1000 Spectrophotometer (NanoDrop Tech., Wilmington, NC, USA) [34]. The DNA specimens were stored at -80 °C and subjected to further analysis utilizing allelic discrimination evaluation.

2.1.4. Analysis of the *CYP1A2* (rs762551; c.-9-154 C>A) variant

The genotyping of the *CYP1A2* (rs762551; c.-9-154 C>A) variant was performed using real-time PCR with a TaqMan allelic discrimination assay, as described previously [26,35]. The reaction protocol was adjusted with a total volume of 25 µl containing 1.25 µl of TaqMan SNP genotyping assay Mix, 12.5 µl TaqMan Universal PCR Master Mix, and 11.25 µl of genomic DNA template. The identified assay for the *CYP1A2* rs762551 variant (C_8881221_40; catalog#: 4362691; Applied Biosystems, Foster City, CA, USA) could detect the transversion substitution at C/A within the target context sequence [VIC/FAM]: TGC TCA AAG GGT GAG CTC TGT GGG C [C/A] CAG GAC GCA TGG TAG ATG GAG CTT A. Furthermore, the PCR amplification process was carried out using the StepOne™ Real-Time PCR System (Applied Biosystems, Foster City, CA, USA). The interpretation of the amplification findings was executed using SDS software version 1.3.1 (Applied Biosystems, Foster City, CA, USA). To avoid the potential false genotyping calls, about 10 % of random specimens were selected and subjected to recycling within independent runs, generating identical consistent findings.

2.1.5. Statistical analysis

The STATA statistical package, version #17 (StataCorp. 2021, College Station, TX: LLC) was used in the processing and tabulation of the data findings. The G*Power, version #3.1.9.7 was used in the computation of the statistical power of this work with an effect size of 0.32, and α error probability of 0.05, thus the proposed study's power was 96 % [36]. The genotypic and allelic frequencies were implemented using the online electronic tool for SNP analysis, SNPStats (<https://www.snpsstats.net/>, accessed on 12 February 2024). Using multiple genetic models, logistic regression analysis were used to find the odds ratio and their corresponding 95 % confidence intervals (CI). These models were adjusted for the women's age, family history, and menopause status [37]. The best-fitting model of inheritance was identified using the Akaike Information Criterion (AIC), which sorted the standard model based on the lowest AIC value [38]. Additionally, the continuous and categorical parameters were measured using the two-sample T-test with equal variances and Chi-square models [39]. A two-tailed *p*-value was conditioned to be below 0.05 to attain a significant value [40].

2.2. Meta-analysis study

2.2.1. Literature screening and search strategy

The design of this established meta-analysis was formulated using the reported checklists for meta-analysis of observational studies in epidemiology [MOOSE] [41]. Moreover, the research findings were manipulated by following the instructions given by the PRISMA-P protocols for systematic reviews and meta-analyses [42]. A systematic search for all available literature was performed pertaining to the involvement of the *CYP1A2* (rs762551; c.-9-154 C>A) genetic variant with the susceptibility to breast carcinoma using various electronic scientific repositories, including PubMed, Scopus, Web of Science, and Science Direct databases, up to February 2024. The prepared strategies used in this work were proposed utilizing multiple relevant keywords including ["Cytochrome P450 Family 1 Subfamily A Member 2", "Cytochrome P450 1A2", "Microsomal Monooxygenase", "P3-450", or "CYP1A2"] and ["rs762551", "CYP1A2*1 F", "c.-163 C>A", or "c.-9-154 C>A"] and ["mutation", "variant", "SNP", "polymorphism", "gene", "genetic", or "genomic"] and ["Breast", "mammary", "neoplasms", "tumors", "carcinoma", "cancer", "BC", or "BRCA"]. Additionally, no restrictions were adjusted for publication year, country of origin, geographic distribution, sample size, or article's language.

2.2.2. Inclusion and exclusion criteria

The selection of eligible records in this meta-analysis was assured based on the subsequent criteria: (1) Observational studies with a case-controlled design; (2) The rationale of selected articles was designed

based on the contribution of the *CYP1A2* (rs762551; c.-9-154 C>A) genetic variant with the susceptibility to breast carcinoma; (3) The target species of the nominated reports were *Homo sapiens* without other species; and (4) Adequate data concerning odds ratio (OR) and their 95 % confidence intervals (95 % CI). Otherwise, the exclusion criteria were settled if one of the following conditions accomplished: (1) The applicable studies did not achieve a "case-controlled design", (2) Repetitive research was detected from previous reports, (3) The pertinent reports were "editorial materials", "abstract", "review articles", or "conference meetings", and (4) Inappropriate allelic/genotypic data were noticed.

2.2.3. Methodological quality assessment and data extraction

The qualified articles were subjected to accurate evaluation to justify and compute the quality score with the aid of two independent researchers using the modified scoring criteria obtained from previous records [43,44]. These amended criteria included the following sections: (1) Representativeness of BRCA patients; (2) The source of healthy controls; (3) Ascertainment of breast carcinoma; (4) Testing for case-control matching; (5) Quality control of genotyping method; (6) Testing for blindness during genotyping examination; (7) Specimen types used for genotyping amplification; (8) Hardy-Weinberg equation (HWE) among healthy controls; (9) Total sample size in selected records; and (10) Assessment of statistical analysis. The calculated scoring ranged from zero (minimum quality) to eighteen (maximum quality) with articles achieving values (≤ 11) being defined as "lowest score", and those attaining values (≥ 12) being regarded as "highest score" (Supplementary Table S1). Additionally, two teams of qualified researchers extracted the relevant information from eligible articles independently to alleviate the risk of publication bias that could arise through the establishment of this work. The processed data was abstracted and incorporated into designed tabulated worksheet containing the following sections: the surname of the first's author, the publication's year, country of origin, geographic distribution, total numbers of sample sizes for BRCA and healthy controls, the appropriate genotyping method used in the amplification protocol, the source of healthy controls ("hospital-based" or "population-based"), the genotypic and allelic frequency of the proposed variant, the *p*-value of calculated HWE among healthy controls, and the quality scoring of particular studies. To avoid any discrepancies in the extracted data, the presented conflicts were manipulated with the aid of a qualified researcher.

2.2.4. Statistical analysis for meta-analysis

Initially, the genotypic/allelic distributions for the *CYP1A2* (rs762551; c.-9-154 C>A) variant were compiled from different articles so that the chi-square method could be used to compute the crude odds ratio and its 95 % confidence intervals [45,46]. This meta-analysis study was conducted using different hereditary models, including allelic, dominant, recessive, homozygote, and heterozygote comparisons [43]. Then, our team performed stratification assessments based on the geographic distribution "ethnicity", genotyping methods, source of healthy controls, and quality scoring of the study. Moreover, the choice of suitable model is a problematic challenge during the establishment of meta-analysis designs [47]. Under the fixed-effect model, the only source of error is the random error within studies, while under the random-effect model, two source of errors could be accounted including within-study and between-study variabilities to widen the range of estimated confidence intervals [48,49]. Thus, the random-effect models could be susceptible to plausible types of publication bias and generate more conservative assessment in the overall estimate compared to fixed-effect model [47]. Due to variability in our subgrouping system, we selected a random-effects model in this design. This model could be employed to calculate heterogeneity between studies utilizing the Q-statistic method with *p*-value ≤ 0.10 and $I^2 > 75\%$ [50,51]. The publication bias was assessed utilizing Egger's regression technique with

p-value > 0.1 and Begg’s funnel plot, which could check for the availability of bias between published studies [44]. The validation of eligible records and the capability of sample sizes to achieve significant findings could be operated using trial sequential analysis (TSA; version 0.9.5.10 beta) [52]. The data analysis for this meta-analysis design was performed using the Comprehensive Meta-analysis software (CMA; version #3.0) [53] and R Studio 2023.12.1 Build 402 with packages “metafor and meta” [54].

3. Results

3.1. Case-control study

3.1.1. Characteristics of the study population

This work comprised 206 female participants that were subdivided into two main groups, including 104 BRCA women with a mean age of 45.7 ± 11.8 years, and 102 healthy controls with a mean age of 46.1 ± 9.46 years (Table 1). Among the BRCA patients, one-third of the females (35.6 %) were classified as having premature menopause, while two-thirds of the females (64.4 %) had late-onset menopause. Moreover, our findings disclosed no significant correlation with demographic data, maternal risk factors, or clinicopathological investigations among the two groups (*p*-value > 0.05).

3.1.2. Functional annotation and bioinformatic analysis

The human cytochrome P450 1 A2 (CYP1A2; ENSG00000140505; other synonyms include P3-450 and CP12) gene is situated on the elongated arm of the chromosome number 15q24.1. This gene spanned 7763 bases (Chr.15: 74,748,845–74,756,607) and aligned along the plus

strand of the chromosome (Fig. 1A). Additionally, the CYP1A2 gene consists of one unique splice transcript with seven exons and six introns. The length of this transcript (CYP1A2–201; ENST00000343932) is 3132 bps and encodes about 516 residues.

The CYP1A2 (rs762551; c.–9–154 C>A) genetic variant is an interesting polymorphism because it is a promoter variant and can be found in the first intron at position 15:74,749,576 of the genomic sequence (Fig. 1B). Furthermore, the protein-protein interaction identified multiple molecular and biological functions, including aflatoxin metabolic processes, xenobiotic catabolic pathways, fatty acids biosynthesis mechanisms, estrogen activity, caffeine oxidase activity, and chemical carcinogenesis (<https://string-db.org/>; Fig. 1C). The crystalline structure of human microsomal P450 1A2 was observed in combination with alpha-naphthoflavone (<https://www.rcsb.org/>; Fig. 1D). The subcellular compartment database identified the availability of the CYP1A2 enzyme within the endoplasmic reticulum membrane (<https://compartments.jensenlab.org/>; Fig. 3E). Moreover, the predicted GeneMANIA database for genetic ontology confirmed the contribution of this potential gene to monooxygenase activity, response to xenobiotic stimulus, steroid metabolic process, and alcohol metabolic process (<https://genemania.org/>; Fig. 1F). The MEXPRESS database looked at the connection between CYP1A2 genetic expression and the initial pathologic diagnosis, the status of the BRCA estrogen receptor and progesterone receptor, histological type, the status of menopause, the stage of the tumor, and the overall survival rate (<https://mexpress.ugent.be/>; Fig. 1G). The pertinent circos plotting was presented using 3DSNP v2.1 (<https://3dsnp.omic.tech/>) to visualize the 3D chromatin configuration and other significant chromatin interactions related to the interest non-coding variant. This circos diagram identified the chromatin loops, histone modifications, transcription factors, and relevant variants (Fig. 1H). Eventually, the target variant’s conservation score was –2.533, calculated from 46 vertebrates and 33 mammals based on genome multiple alignments (Fig. 1I).

Table 1
The demographic, maternal, and clinicopathological parameters of the study population.

Variables	Levels	BRCA women (n=104)	Healthy controls (n=102)	P- value
I. Demographic data and maternal characteristics				
1. Age, years	Mean ± SD	45.7 ± 11.8	46.1 ± 9.46	0.852
2. Age groups	< 35 years, n (%)	13 (12.5)	13 (12.7)	0.379
	35–54 years, n (%)	77 (74.0)	68 (66.7)	
	≥ 55 years, n (%)	14 (13.5)	21 (20.6)	
3. Gender	Female, n (%)	104 (100.0)	102 (100.0)	NA
4. Family history	Positive, n (%)	25 (24.0)	—	NA
	Negative, n (%)	79 (76.0)	—	
5. Menopause status	Premature menopause, n (%)	37 (35.6)	27 (26.5)	0.158
	Late-onset menopause, n (%)	67 (64.4)	75 (73.5)	
	II. Clinicopathological investigations			
1. Tumor grade	Grade I, n (%)	33 (31.7)	—	NA
	Grade II, n (%)	56 (53.9)	—	
	Grade III, n (%)	15 (14.4)	—	
2. T stage	T1, n (%)	44 (42.3)	—	NA
	T2, n (%)	40 (38.5)	—	
	T3, n (%)	20 (19.2)	—	
3. N stage	N1, n (%)	47 (45.2)	—	NA
	N2, n (%)	44 (42.3)	—	
	N3, n (%)	13 (12.5)	—	
4. M stage	M0, n (%)	46 (44.2)	—	NA
	M1, n (%)	58 (55.8)	—	
5. ER/PR status	Positive, n (%)	28 (26.9)	—	NA
	Negative, n (%)	76 (73.1)	—	

Data are expressed as numbers (%) or mean ± SD. The Chi-square test and two-sample T-test with equal variances were applied. **Abbreviations:** BRCA: Breast carcinoma; SD: Standard deviation; ER/PR: Estrogen/Progesterone receptors. Bold values indicate the *P*-value < 0.05.

Note: Premature menopause is a term that describes a woman which began to menopause prior to the age of 45 years, while late-onset menopause stands for a woman didn’t introduce menopause at 55 years or older.

3.1.3. Genetic analysis of the CYP1A2*rs762551 variant

Interestingly, the genotypic and allelic distributions for the CYP1A2 (rs762551; c.–9–154 C>A) variant among BRCA women compared to healthy controls were illustrated in (Table 2). The most prominent genotype in the study population was heterozygosity, with 44.2 % compared to other genotypes.

Furthermore, the wild type (C/C genotype) was observed in 27.2 % of the whole participants, 37.5 % of BRCA women, and 16.7 % of healthy controls. The heterozygous variant (C/A genotype) was the most frequent polymorphic variant among BRCA women, with 41.3 % compared with 47.0 % among healthy controls (OR = 0.39, *p*-value = 0.008). The mutant homozygous (A/A genotype) was identified in 21.2 % of BRCA patients compared to 36.3 % of healthy controls (OR = 0.26, *p*-value < 0.001). According to the Hardy-Weinberg equilibrium (HWE), the observed and expected values of BRCA subjects and healthy controls reached equilibrium with this proposed equation (*p*-value > 0.05). Regarding allelic frequencies, the common C allele was observed within 49.3 % of the entire subjects, with being the most prevalent among BRCA women with 58.2 % compared to 40.2 % among control cohorts. On the other hand, the rare A allele was identified within 50.7 % of the whole participants, 41.8 % among BRCA patients, and 59.8 % among healthy controls. Significantly, the rare allelic variant (rs762551*A allele) was associated with protection against the development of breast carcinoma (OR = 0.48, *p*-value < 0.001).

3.1.4. The potential impact of the CYP1A2*rs762551 variant with breast carcinoma

The crude and adjusted hereditary association models for the CYP1A2*rs762551 polymorphism with susceptibility to BRCA were explained in (Table 3). Our findings confirmed the association of the CYP1A2*rs762551 variant with protection against the progression of breast carcinoma. Interestingly, this intronic variant was associated with

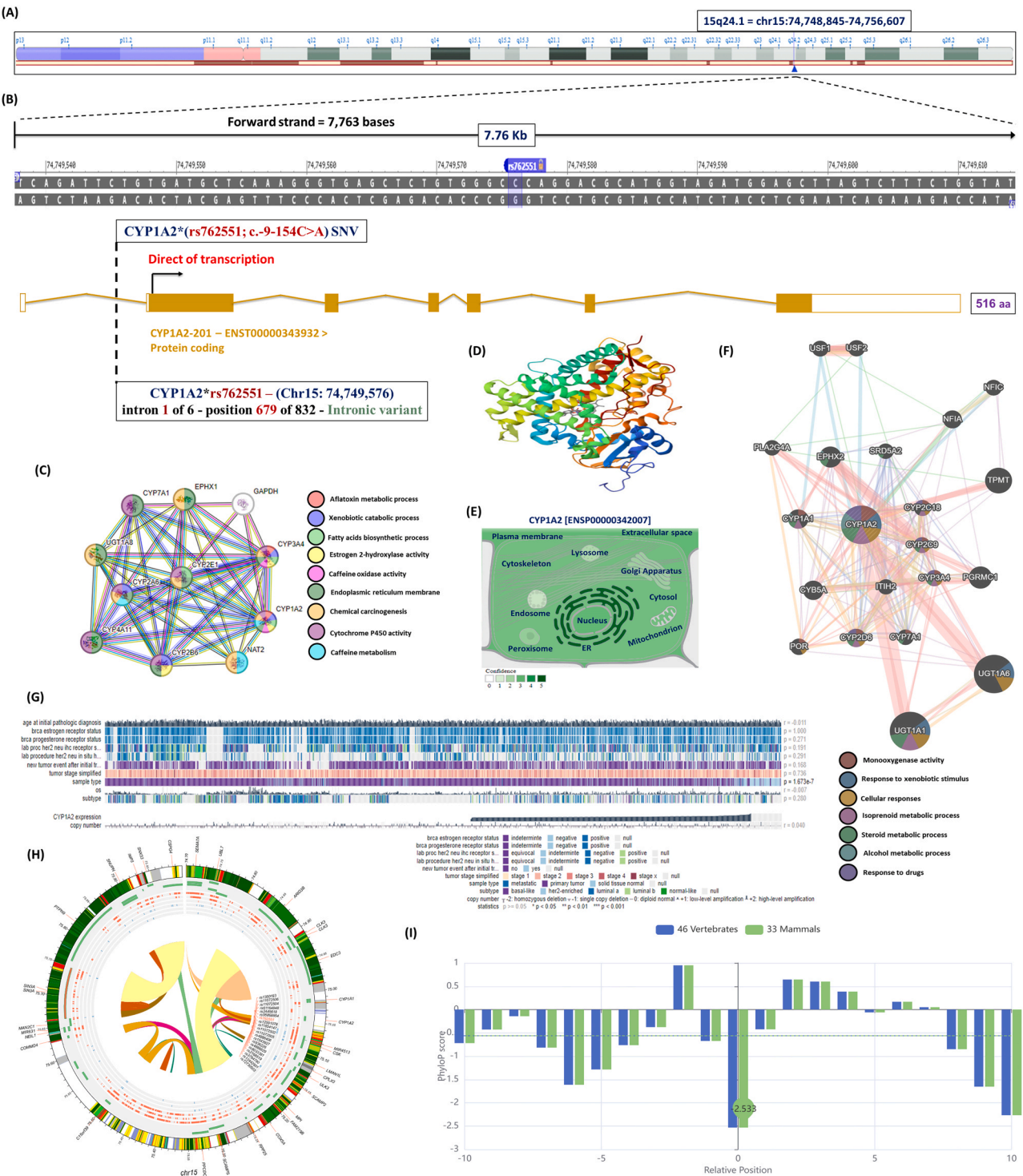


Fig. 1. Bioinformatic framework analysis for *CYP1A2**(rs762551; c.-3-154 C>A) gene variant. (A) The chromosomal localization of the *CYP1A2* gene. It is positioned on chromosome # 15q24.1 and traverses about 7763 bases [Chr.15: 74,748,845–74,756,607] along the plus strand (NCBI Gene Database; GRCh38). (B) The genomic structure of the *CYP1A2**(rs762551; c.-3-154 C>A) variant. This gene is transcribed into a single splice variant with seven exons and six introns. The length of this transcript (*CYP1A2*-201; ENST00000343932) is 3132 bps and encoding about 516 residues. The *CYP1A2* (rs762551; c.-3-154 C>A) variant is considered as a promoter variant and located in the first intron at position 15:74,749,576 of the genomic sequence (<https://www.ensembl.org/>). (C) The protein-protein interaction (PPI) of the *CYP1A2* protein using the STRING database (<https://string-db.org/>). (D) The crystalline structure of human *CYP1A2* protein complexed with alpha-naphthoflavone using Protein Data Bank (<https://www.rcsb.org/>). (E) The subcellular localization of the *CYP1A2* protein with darker colors representing excess abundance (<https://compartments.jensenlab.org/>). (F) The gene-gene interaction network for the *CYP1A2* gene using GeneMania database (<https://genemania.org/>). (G) The MEXPRESS database studied the linkage between the *CYP1A2* gene expression and initial pathologic diagnosis, BRCA estrogen receptor status, BRCA progesterone receptor status, histological type, menopause status, tumor stage, and overall survival (<https://mexpres.ugent.be/>). (H) Circos plot representing the chromatin loops and other signatures of the *CYP1A2* (rs762551; c.-3-154 C>A) variant using 3DSNP v2.1 (<https://3dsnp.omic.tech/>; accessed on 29 February 2024). (I) The conservation score of the variant of interest is -2.533, calculated from 46 vertebrates and 33 mammals based on genome multiple alignments.

Table 2
The genotypic, and allelic frequencies of the *CYP1A2* (*rs762551*; *c.-9-154 C>A*) variant.

Genetic polymorphisms	All subjects	BRCA women	Healthy controls	OR (95 % CI)	P-value
Genotypic frequencies	n (%)	n (%)	n (%)		
C/C	206 (27.2)	104 (37.5)	102 (16.7)	1.0	
C/A	91 (44.2)	43 (41.3)	48 (47.0)	0.39 (0.19–0.79)	0.008
A/A	59 (28.6)	22 (21.2)	37 (36.3)	0.26 (0.12–0.56)	< 0.001
HWE	$\chi^2=2.79$, $p=0.095$	$\chi^2=2.35$, $p=0.125$	$\chi^2=0.04$, $p=0.830$		
Allelic frequencies	n (%)	n (%)	n (%)		
(C) allele	412 (49.3)	208 (58.2)	204 (40.2)	1.0	
(A) allele	209 (50.7)	87 (41.8)	122 (59.8)	0.48 (0.33–0.71)	< 0.001

Data are expressed as numbers with percentages. The Chi-square test was applied.

Abbreviations: BRCA: Breast carcinoma; HWE, Hardy–Weinberg equilibrium; OR: Odds Ratio; CI: Confidence Intervals. Bold values indicate the *P*-value < 0.05.

a decreased risk of breast carcinoma compared to healthy controls under the dominant (OR = 0.34, *p*-value < 0.001), recessive (OR = 0.44, *p*-value = 0.011), and homozygous (OR = 0.25, *p*-value < 0.001) models, as described in (Table 3). According to the Akaike Information Criterion (AIC), the best selection model of inheritance was the dominant model.

3.1.5. Interaction analysis of the *CYP1A2***rs762551* variant and menopause status

As illustrated in Table 4, our findings revealed no significant association between this potential variant and the risk of breast carcinoma

among women with premature menopause under different genetic models (*p*-value > 0.05). Conversely, women with late-onset menopause and carrying the A allele indicated protection against the development of breast carcinoma under dominant (OR = 0.33, *p*-value = 0.004), recessive (OR = 0.45, *p*-value = 0.042), and homozygous (OR = 0.25, *p*-value = 0.001) models, as mentioned in (Table 4).

3.2. Meta-analysis study

3.2.1. Literature search screening

Initially, a total of 2006 retrieved records were obtained from available electronic repositories, including: [PubMed = 510, Web of Science = 134, Scopus = 879, and Science Direct = 483]. After precise evaluation and removal of duplicated articles, only 574 records were subjected to additional screening using their titles and abstracts (Fig. 2). Then, 540 records were eliminated due to various reasons that were attributed to review records (n = 178), no case-controlled articles (n = 253), editorial materials (n = 28), conference abstracts (n = 69), and meta-analysis reports (n = 12). The remaining 34 articles were assessed for eligibility with removal of 18 studies due to other conditions. Ultimately, a sum of 17 eligible articles [16 retrieved records + the present study] were assembled and met the inclusion criteria for the association of the *CYP1A2* (*rs762551*; *c.-9-154 C>A*) variant with the susceptibility to breast carcinoma.

3.2.2. Characteristics of the retrieved articles

The designated characteristics of the seventeen eligible articles were displayed in (Table 5) among BRCA women and healthy controls. This meta-analysis design included 12922 BRCA patients and 15603 healthy controls. Four reports were performed among American subjects [55–58], four articles [three articles + the current work] were done among Arabian subjects [10,21,26], five studies were carried out among Caucasian subjects [27,59–62], three articles were done among Asian subjects [28,63,64], and one article was done among Brazilian subjects

Table 3
Genetic association models of the *CYP1A2* (*rs762551*; *c.-9-154 C>A*) variant with the BRCA susceptibility.

Genetic association models		BRCA women	Healthy controls	Statistical analysis					
Model	Genotypes	n (%) 104	n (%) 102	Crude OR (95 % CI)	P-value	AIC	Adjusted OR (95 % CI)	P-value	AIC
Codominant	C/C	39 (37.5)	17 (16.7)	1.0	0.002	278.6	1.0	< 0.001	247.8
	C/A	43 (41.3)	48 (47.0)	0.39 (0.19–0.79)			0.40 (0.20–0.82)		
	A/A	22 (21.2)	37 (36.3)	0.26 (0.12–0.56)			0.25 (0.11–0.55)		
Dominant	C/C	39 (37.5)	17 (16.7)	1.0	< 0.001	278.0	1.0	< 0.001	247.1
	C/A + A/A	65 (62.5)	85 (83.3)	0.33 (0.17–0.64)			0.34 (0.17–0.65)		
Recessive	C/C + C/A	82 (78.8)	65 (63.7)	1.0	0.016	283.8	1.0	0.011	251.3
	A/A	22 (21.2)	37 (36.3)	0.47 (0.25–0.88)			0.44 (0.24–0.83)		

Data are expressed as numbers with percentages. Adjusted by age, family history, and menopause status. The chi-square test was applied. **Abbreviations:** BRCA: Breast carcinoma; OR: Odds Ratio; CI: Confidence Intervals; AIC: Akaike information criterion. Bold values indicate the *P*-value < 0.05.

Table 4
Interaction analysis between the *CYP1A2* (*rs762551*; *c.-9-154 C>A*) variant and menopause status.

Genetic association models		Menopause status							
		Premature menopause				Late-onset menopause			
		BRCA women	Healthy controls	Statistical analysis		BRCA women	Healthy controls	Statistical analysis	
Model	Genotypes	n (%) 37	n (%) 27	Crude OR (95 % CI)	P-value	n (%) 67	n (%) 75	Crude OR (95 % CI)	P-value
Codominant	C/C	13 (35.1)	4 (14.8)	1.0	0.086	26 (38.8)	13 (17.3)	1.0	0.001
	C/A	15 (40.6)	12 (44.5)	0.38 (0.10–1.49)		28 (41.8)	36 (48.0)	0.39 (0.17–0.89)	
	A/A	9 (24.3)	11 (40.7)	0.25 (0.06–1.05)		13 (19.4)	26 (34.7)	0.25 (0.10–0.64)	
Dominant	C/C	13 (35.1)	4 (14.8)	1.0	0.069	26 (38.8)	13 (17.3)	1.0	0.004
	C/A + A/A	24 (64.9)	23 (85.2)	0.32 (0.09–1.13)		41 (61.2)	62 (82.7)	0.33 (0.15–0.72)	
Recessive	C/C + C/A	28 (75.7)	16 (59.3)	1.0	0.162	54 (80.6)	49 (65.3)	1.0	0.042
	A/A	9 (24.3)	11 (40.7)	0.47 (0.16–1.37)		13 (19.4)	26 (34.7)	0.45 (0.21–0.98)	

Data are expressed as numbers with percentages. The chi-square test was applied. **Abbreviations:** BRCA: Breast carcinoma; OR: Odds Ratio; CI: Confidence Intervals. Bold values indicate the *P*-value < 0.05.

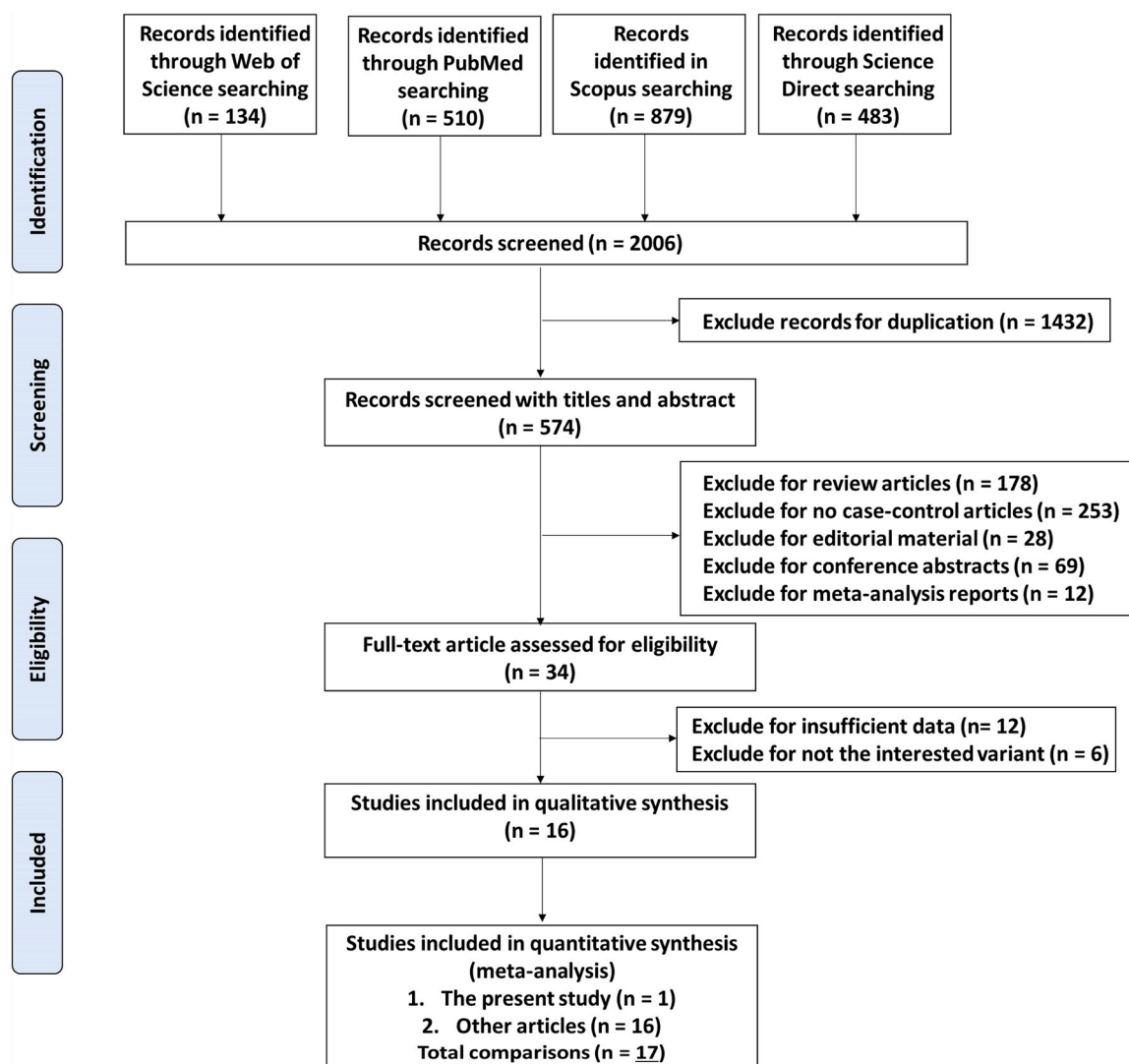


Fig. 2. The flowchart diagram of the study selection procedure.

[65]. Additionally, these studies were genotyped using accredited genotyping methods, including PCR-RFLP, TaqMan SNP genotyping assay, Sequenom MassARRAY system, Illumina GoldenGate assay, Sentrix HumanHap550 genotyping assay, and MALDI-TOF MS. Moreover, nine articles were hospital-based, while the remaining eight records were population-based. In terms of HWE among healthy controls, sixteen studies were in compliance with this equation, and only one record did not achieve equilibrium [10]. Lastly, seven articles revealed the lowest quality score ($QS \leq 11$), while ten articles attained the highest quality score ($QS \geq 12$), (Supplementary Table S2).

3.2.3. Meta-analysis of *CYP1A2**(rs762551) variant and susceptibility to BRCA

This meta-analysis study was performed to investigate the contribution of the *CYP1A2* (rs762551; c.-9-154 C>A) variant with the susceptibility to breast carcinoma among different ethnic subjects. Our team studied the correlation of this intronic variant with increased risk of breast carcinoma among the overall subjects, then we performed subgroup stratification based on geographical distribution “ethnicity”, genotyping techniques, source of healthy controls, and quality assessment. This meta-analysis design did not attain statistical significance with the *CYP1A2**(rs762551) variant and elevated risk of breast carcinoma among the pooled subjects under the allelic [OR = 0.938, 95 % CI = 0.854–1.031, P -value = 0.183], dominant [OR = 0.936, 95 % CI =

0.782–1.119, P -value = 0.465], recessive [OR = 0.955, 95 % CI = 0.859–1.062, P -value = 0.395], homozygote [OR = 0.930, 95 % CI = 0.750–1.152, P -value = 0.506], and heterozygote [OR = 0.955, 95 % CI = 0.805–1.133, P -value = 0.597] models, as being shown in (Fig. 3).

Upon stratification by geographical distribution “ethnicity”, our findings revealed noticeable associations with the *CYP1A2**(rs762551) variant and protection against the development of breast carcinoma among Caucasian subjects under the allelic [OR = 0.751, 95 % CI = 0.584–0.965, P -value = 0.025], dominant [OR = 0.584, 95 % CI = 0.378–0.901, P -value = 0.015], homozygote [OR = 0.541, 95 % CI = 0.392–0.888, P -value = 0.015], and heterozygote [OR = 0.636, 95 % CI = 0.437–0.926, P -value = 0.018] models. Interestingly, this work disclosed a significant association with this promoter variant and increased risk of breast carcinoma among other ethnicities (Asian and Brazilian subjects) under the dominant [OR = 1.161, P -value = 0.033] model, as being represented in (Fig. 4). Conversely, our findings failed to achieve a significant difference for the *CYP1A2* (rs762551; c.-9-154 C>A) variant and the susceptibility to breast carcinoma under various genetic models stratified by genotyping methods, source of healthy controls, and quality scoring (p -value > 0.05, Table 6).

3.2.4. Heterogeneity analysis and publication bias

To minimize the publication bias and heterogeneity between studies, our team conducted this meta-analysis design using the random-effects

Table 5
Characteristics of the studies for the CYP1A2 (rs762551; c.-9-154 C>A) variant included in the meta-analysis.

First Author	Year	Country	Geographic distribution	Numbers		Genotyping method	Source of controls	BRCA/ Controls (Genotype)			BRCA/ Controls (Allele)		P (HWE) Controls	QS
				BRCA	Controls			CC	CA	AA	C allele	A allele		
The present study	2024	Egypt	Arabian	104	102	TaqMan SNP Genotyping assay	PB	39/17	43/48	22/37	121/82	87/122	0.125	11
Al-Eitan	2019	Jordan	Arabian	220	217	Sequenom MassARRAY® system	PB	23/38	99/84	98/95	145/160	295/274	0.013	7.5
Parada	2017	USA	American	975	1017	Illumina GoldenGate assay	PB	82/90	422/417	471/510	586/597	1364/1437	0.718	14
Sun	2015	China	Others	530	546	Sequenom MassARRAY® system	PB	52/57	230/250	248/239	334/364	726/728	0.481	13
Imene	2015	Tunisia	Arabian	18	27	TaqMan SNP Genotyping assay	PB	2/4	11/14	5/9	15/22	21/32	0.701	11
Lowcock	2013	Canada	Caucasian	1693	1761	Sequenom MassARRAY® system	HB	161/138	656/737	876/886	978/1013	2408/2509	0.372	12
Lee	2013	USA	American	579	981	Sentrix HumanHap550 genotyping assay	HB	56/79	220/388	303/514	332/546	826/1416	0.631	14
Ayari	2013	Tunisia	Arabian	117	42	TaqMan SNP Genotyping assay	PB	17/4	58/20	42/18	92/28	142/56	0.643	13
Khvostova	2012	Russia	Caucasian	323	526	PCR-RFLP	PB	40/38	148/220	135/268	228/296	418/756	0.432	7
Marie-Genica	2010	German	Caucasian	3147	5485	MALDI-TOF MS	PB	265/447	1241/2262	1641/2776	1771/3156	4523/7814	0.645	16
Shimada	2009	Brazil	Others	872	873	TaqMan SNP Genotyping Assay	PB	96/120	414/383	362/370	606/623	1138/1123	0.192	12
Sangrajrang	2009	Thailand	Others	552	483	TaqMan SNP Genotyping Assay	PB	325/301	180/166	47/16	830/768	274/198	0.231	13.5
Gulyaeva	2008	Russia	Caucasian	93	180	PCR-RFLP	HB	22/2	38/58	33/120	82/62	104/298	0.081	6
Takata	2007	USA	American	1108	614	PCR-RFLP	PB	63/46	497/253	548/315	623/345	1593/883	0.623	12
Kotsopoulos	2007	Canada	Caucasian	170	241	PCR-RFLP	HB	17/12	68/107	85/122	102/131	238/351	0.059	9
Long	2006	China	Others	1082	1139	PCR-RFLP	PB	157/182	567/576	358/381	881/940	1283/1338	0.144	15
Le Marchand	2005	USA	American	1339	1369	PCR-RFLP	PB	91/125	608/627	640/617	790/877	1888/1861	0.055	11

Abbreviations: BRCA: Breast carcinoma; PCR-RFLP: Polymerase chain reaction-restriction fragment length polymorphism; MALDI-TOF MS: Matrix assisted laser desorption ionization-time of flight mass spectrometry; PB: Population-based; HB: Hospital-based; HWE: Hardy–Weinberg equilibrium of control; QS: Quality score.

model, as illustrated in (Table 6). Additionally, Begg’s funnel plot was conceived to test the symmetric manner between published records under multiple hereditary models, (Fig. 5). However, eligible articles for the CYP1A2*rs762551 variant among BRCA women showed a degree of publication bias under the allelic model (*P-value* for Egger’s regression test = 0.053), as depicted in (Table 6). Thus, we performed the trim and fill method to validate the observed results, with no changes within the adjusted ones.

3.2.5. Trial sequential analysis (TSA)

Our team used trial sequential analysis to validate the selected sample sizes and verify these values were suitable to obtain significant findings [43,44]. The accessible data for this interest variant among BRCA women was selected under the dominant model with a total sample size of 28525 participants. Essentially, the cumulative Z-curve did not transverse the observed boundaries before achieving the necessary sample sizes, and thus, the validity of these findings required additional records to authenticate the hypothesis, as noticed in (Fig. 5).

4. Discussion

The cytochrome P450 1A2 (CYP1A2) gene belongs to the cytochrome oxidase lineage that encodes a definite enzyme and implicates multiple processes involving the conversion of pro-carcinogens into mutagenic metabolites [23,66]. These reactive metabolites could engage with various genetic materials (DNA and RNA), leading to the production of adduct compounds that trigger carcinogenesis progression, particularly breast carcinoma [16,29]. Consequently, our team was motivated to investigate the association of this promoter variant with the susceptibility to breast carcinoma. This case-control research and meta-analysis design were performed based on 17 eligible articles to disclose the potential impact of the CYP1A2*rs762551 variant with BRCA progression. Interestingly, our findings revealed the correlation of this intronic variant with a decreased risk of breast carcinoma among Egyptian women. The CYP1A2*rs762551 variant conferred protection against cancer development under allelic (OR = 0.48, and *p-value* < 0.001), dominant (OR = 0.34, and *p-value* < 0.001), and recessive (OR =

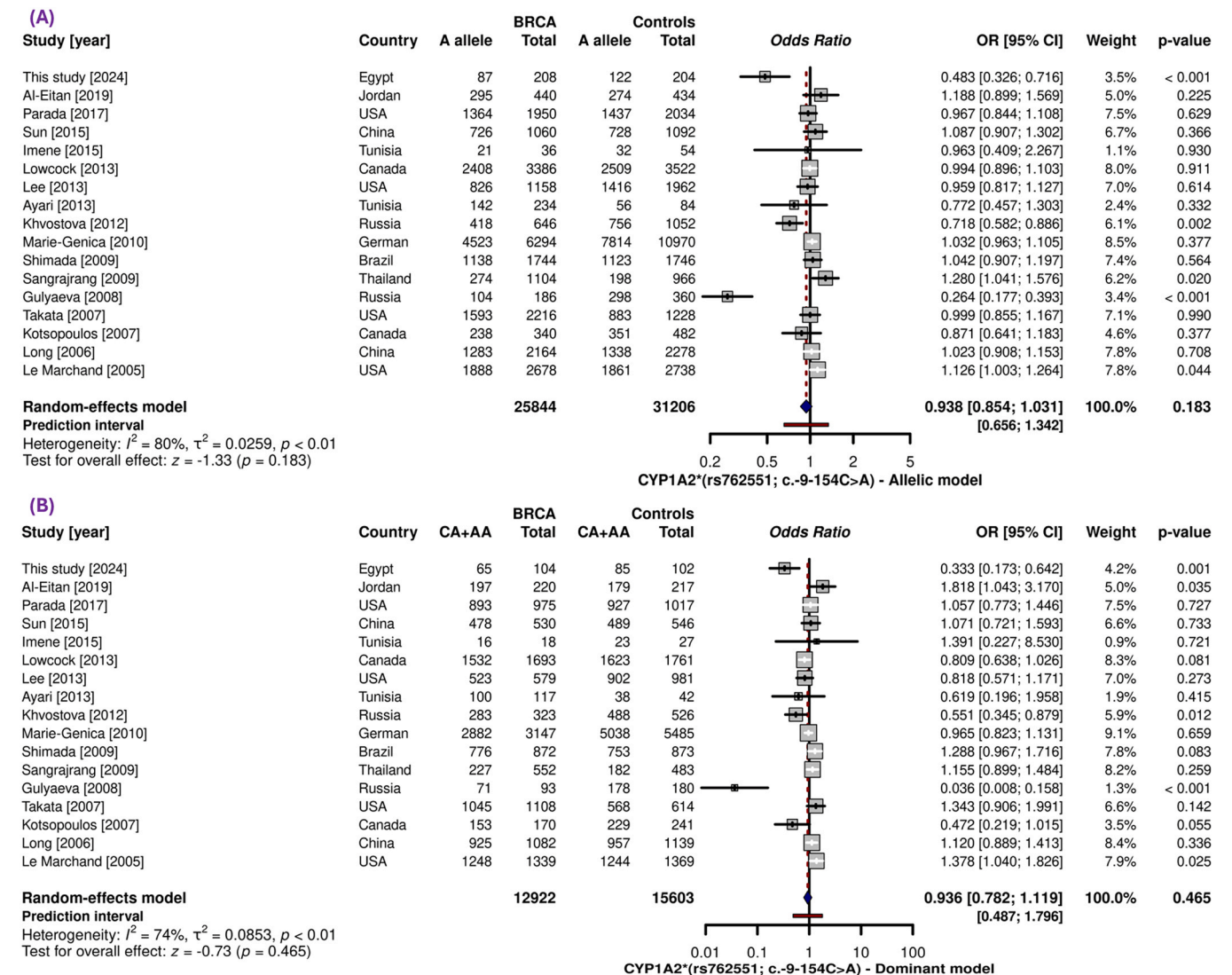


Fig. 3. Forest plot of the pooled data for the association of the *CYP1A2**(rs762551; c.-9-154 C>A) variant and the susceptibility to breast carcinoma among the overall populations under (A): the allelic model; and (B): the dominant model.

0.44, and p -value = 0.011) models. Moreover, BRCA women with late-onset menopause observed 55 % decreased risk against cancer development with the *CYP1A2**rs762551 A/A genotype, compared to other groups. Consistent with our findings, we observed two reports among Russian females conferring decreased risk with the *CYP1A2**rs762551 variant and BRCA progression under dominant and recessive models [59,61]. Otherwise, the *CYP1A2**rs762551 (A allele) was associated with an elevated risk of BRCA among Thai and American women (p -value = 0.020 and 0.044) [28,58]. This promoter variant was correlated with an increased risk of breast carcinoma among Jordanian women under the dominant (OR = 1.82, and p -value = 0.035) model [10]. Other reports unveiled contrasting findings with no significant association between the *CYP1A2**rs762551 variant and the susceptibility to breast carcinoma among various ethnic women, including American [55-57], Chinese [63,64], Tunisian [21,26], Canadian [27,62], German [60], and Brazilian [65] subjects. Thus, this work revealed the association of the *CYP1A2**rs762551 variant and protection against BRCA risk among Egyptian females.

Alternatively, this meta-analysis design did not achieve significant differences with the *CYP1A2**rs762551 variant and increased risk of breast carcinoma among the overall populations using different genetic models. Conversely, this promoter variant revealed a significant

association with protection against BRCA development among Caucasian subjects. Interestingly, the *CYP1A2* (rs762551; c.-9-154 C>A) variant has higher fluctuations with enzymatic activity leading to cancer development, but with inadequate assumptions [23,58]. After precise literature screening, we encountered four meta-analysis records studying the association of this potential variant with the susceptibility to BRCA among various ethnic populations [13,16,29,67]. In concordance with our findings, our team observed a meta-analysis study revealing a significant difference between the *CYP1A2**rs762551 (A allele) and protection against cancer risk (OR = 0.92, and p -value = 0.013). Moreover, they identified a strong association with this allelic variant and an elevated risk of breast carcinoma among the overall subjects (OR = 1.05, and p -value = 0.024) [29]. Other meta-analysis records identified the contribution of the *CYP1A2**rs762551 (C allele) to the increased risk of breast carcinoma among Caucasian subjects under different genetic models [16,67]. Conversely, another meta-analysis study indicated no significant association with the *CYP1A2**rs762551 variant and the susceptibility to breast carcinoma among Caucasian and Asian women (OR = 1.71 and 1.27, respectively) [13]. In this respect, we could speculate that the *CYP1A2* (rs762551; c.-9-154 C>A) variant was correlated with a decreased risk of breast carcinoma among Caucasian women.

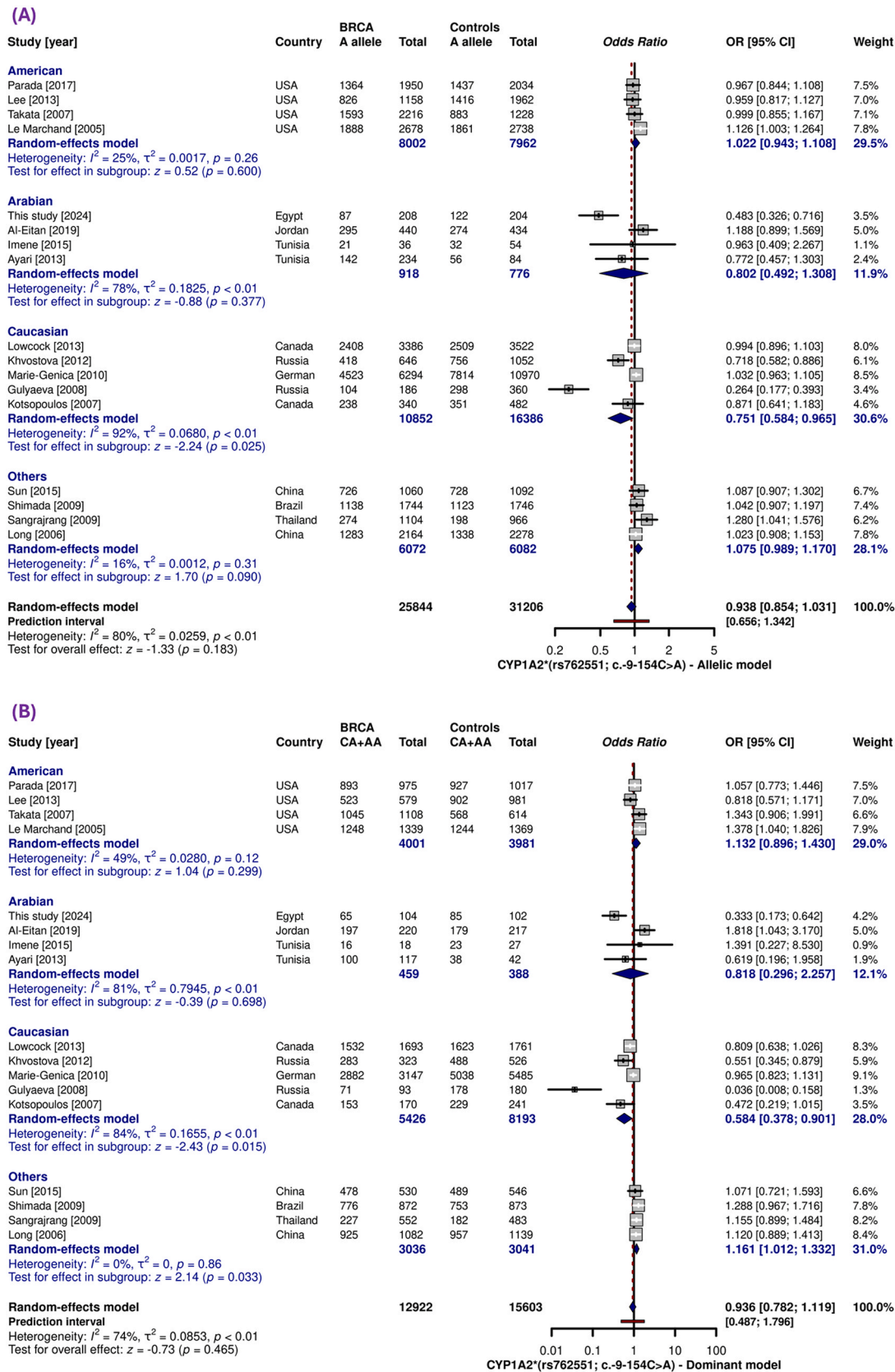


Fig. 4. Forest plot of the pooled data for the association of the *CYP1A2**(rs762551; c.-9-154 C>A) variant and the susceptibility to breast carcinoma stratified by geographical distribution under (A): the allelic model; and (B): the dominant model.

Table 6

Meta-analysis of the association between CYP1A2 (rs762551; c.-9-154 C>A) variant and BRCA susceptibility.

Comparison	Subgroups	No. of studies	Sample size		Test of association				Test of heterogeneity			Publication bias
			BRCA	Controls	OR	95 % CI	P-value	Weight	Q test	P-value	I ² (%)	P-value (Egger's)
Allelic model (A allele versus C allele)	Overall	17	25,844	31,206	0.938	0.854–1.031	0.183	100 %	80.99	< 0.001	80.24	0.053
	Ethnicity											
	Arabian	4	918	776	0.802	0.492–1.308	0.377	11.9 %	13.67	0.003	78.05	0.666
	Caucasian	5	10,852	16,386	0.751	0.584–0.965	0.025	30.6 %	52.69	< 0.001	92.41	0.071
	American	4	8002	7962	1.022	0.943–1.108	0.600	29.5 %	4.011	0.260	25.20	0.157
	Other ethnicities	4	6072	6082	1.075	0.989–1.170	0.090	28.1 %	3.587	0.310	16.35	0.089
	Genotyping method											
	PCR-RFLP	6	8230	8138	0.801	0.628–1.023	0.075	36.8 %	56.73	< 0.001	91.19	0.027
	TaqMan genotyping	5	3326	3054	0.891	0.653–1.215	0.466	20.5 %	19.70	< 0.001	79.69	0.390
	Other methods	6	14,288	20,014	1.017	0.970–1.067	0.491	42.6 %	3.095	0.685	0.0	0.698
	Source of controls											
	Population-based	8	20,080	24,816	1.021	0.980–1.064	0.326	56.1 %	5.468	0.603	0.0	0.178
	Hospital-based	9	5764	6390	0.816	0.632–1.053	0.118	43.9 %	72.50	< 0.001	88.96	0.160
	Quality score											
Dominant model (C/A+A/A versus C/C)	Highest quality	10	21,310	25,882	1.022	0.982–1.064	0.282	68.5 %	7.810	0.553	0.0	0.925
	Lowest quality	7	4534	5324	0.726	0.512–1.031	0.073	31.5 %	68.61	< 0.001	91.25	0.229
	Overall	17	12,922	15,603	0.936	0.782–1.119	0.465	100 %	62.45	< 0.001	74.38	0.113
	Ethnicity											
	Arabian	4	459	388	0.818	0.296–2.257	0.698	12.1 %	15.59	< 0.001	80.76	0.872
	Caucasian	5	5426	8193	0.584	0.378–0.901	0.015	28.0 %	25.79	< 0.001	84.49	0.012
	American	4	4001	3981	1.132	0.896–1.430	0.299	29.0 %	5.921	0.116	49.33	0.651
	Other ethnicities	4	3036	3041	1.161	1.012–1.332	0.033	31.0 %	0.755	0.860	0.0	0.898
	Genotyping method											
	PCR-RFLP	6	4115	4069	0.740	0.461–1.188	0.213	33.5 %	36.92	< 0.001	86.46	0.031
	TaqMan genotyping	5	1663	1527	0.873	0.552–1.382	0.562	23.0 %	15.08	0.005	73.48	0.361
	Other methods	6	7144	10,007	0.974	0.831–1.141	0.745	43.5 %	8.482	0.132	41.05	0.335
	Source of controls											
	Population-based	8	10,040	12,408	1.027	0.891–1.183	0.714	56.7 %	13.23	0.067	47.09	0.921
Recessive model (A/A versus C/C+ C/A)	Hospital-based	9	2882	3195	0.738	0.485–1.123	0.156	43.3 %	49.18	< 0.001	83.73	0.078
	Quality score											
	Highest quality	10	10,655	12,941	1.033	0.929–1.150	0.547	71.4 %	12.40	0.192	27.40	0.875
	Lowest quality	7	2267	2662	0.587	0.298–1.157	0.124	28.6 %	49.17	< 0.001	87.80	0.144
	Overall	17	12,922	15,603	0.955	0.859–1.062	0.395	100 %	54.60	< 0.001	70.69	0.131
	Ethnicity											
	Arabian	4	459	388	0.765	0.514–1.138	0.185	9.30 %	4.558	0.207	34.17	0.483
	Caucasian	5	5426	8193	0.808	0.624–1.048	0.108	32.5 %	32.42	< 0.001	87.66	0.095
	American	4	4001	3981	1.002	0.914–1.100	0.958	32.5 %	3.230	0.358	7.113	0.276
	Other ethnicities	4	3036	3041	1.148	0.897–1.468	0.273	25.7 %	11.91	0.008	74.81	0.007
	Genotyping method											
	PCR-RFLP	6	4115	4069	0.818	0.641–1.045	0.108	38.3 %	30.88	< 0.001	83.81	0.045
	TaqMan genotyping	5	1663	1527	0.966	0.564–1.655	0.899	15.3 %	17.89	< 0.001	77.64	0.966
	Other methods	6	7144	10,007	1.042	0.981–1.108	0.183	46.5 %	2.509	0.775	0.0	0.613

(continued on next page)

Table 6 (continued)

Comparison	Subgroups	No. of studies	Sample size		Test of association				Test of heterogeneity			Publication bias
			BRCA	Controls	OR	95 % CI	P-value	Weight	Q test	P-value	I ² (%)	P-value (Egger's)
Homozygous model (A/A versus C/C)	Source of controls											
	Population-based	8	10,040	12,408	1.031	0.977–1.088	0.265	61.6 %	5.253	0.629	0.0	0.054
	Hospital-based	9	2882	3195	0.857	0.635–1.156	0.311	38.4 %	45.63	< 0.001	82.47	0.600
	Quality score											
	Highest quality	10	10,655	12,941	1.023	0.944–1.109	0.576	70.3 %	15.89	0.069	43.36	0.720
	Lowest quality	7	2267	2662	0.722	0.510–1.023	0.067	29.7 %	35.53	< 0.001	83.11	0.119
	Overall	17	7422	8993	0.930	0.750–1.152	0.506	100 %	73.98	< 0.001	78.37	0.194
	Ethnicity											
	Arabian	4	248	222	0.701	0.233–2.108	0.527	12.9 %	14.87	0.002	79.82	0.727
	Caucasian	5	3275	4809	0.541	0.329–0.888	0.015	29.1 %	32.01	< 0.001	87.50	0.017
	American	4	2254	2296	1.122	0.883–1.425	0.347	29.7 %	5.802	0.122	48.29	0.552
	Other ethnicities	4	1645	1666	1.317	0.977–1.775	0.071	28.3 %	7.980	0.046	62.40	0.158
	Genotyping method											
	PCR-RFLP	6	2189	2228	0.674	0.396–1.146	0.145	35.0 %	42.41	< 0.001	88.21	0.026
	TaqMan genotyping	5	957	896	0.910	0.404–2.049	0.820	20.7 %	23.93	< 0.001	83.28	0.602
	Other methods	6	4276	5869	0.981	0.859–1.120	0.776	44.2 %	6.055	0.301	17.43	0.396
	Source of controls											
Heterozygous model (C/A versus C/C)	Population-based	8	5771	7128	1.030	0.902–1.176	0.659	57.2 %	10.83	0.146	35.34	0.875
	Hospital-based	9	1651	1865	0.702	0.397–1.242	0.224	42.8 %	62.95	< 0.001	87.29	0.183
	Quality score											
	Highest quality	10	6170	7489	1.070	0.922–1.241	0.373	69.5 %	17.97	0.036	49.91	0.399
	Lowest quality	7	1252	1504	0.510	0.239–1.090	0.083	30.5 %	54.57	< 0.001	89.01	0.111
	Overall	17	7008	8310	0.955	0.805–1.133	0.597	100 %	51.53	< 0.001	68.95	0.251
	Ethnicity											
	Arabian	4	292	229	0.912	0.350–2.376	0.850	10.9 %	12.31	0.006	75.64	0.869
	Caucasian	5	2656	4021	0.636	0.437–0.926	0.018	28.2 %	17.19	0.002	76.73	0.013
	American	4	2039	2025	1.148	0.904–1.457	0.258	29.2 %	5.631	0.131	46.73	0.719
	Other ethnicities	4	2021	2035	1.124	0.973–1.299	0.111	31.6 %	2.406	0.492	0.0	0.925
	Genotyping method											
	PCR-RFLP	6	2316	2246	0.824	0.537–1.263	0.373	33.4 %	27.31	< 0.001	81.69	0.029
	TaqMan genotyping	5	1185	1077	0.915	0.605–1.383	0.673	22.7 %	11.07	0.026	63.86	0.496
	Other methods	6	3507	4987	0.959	0.797–1.154	0.658	43.9 %	10.22	0.069	51.09	0.293
	Source of controls											
	Population-based	8	5161	6391	1.021	0.870–1.198	0.801	57.9 %	15.23	0.033	54.04	0.729
	Hospital-based	9	1847	1919	0.796	0.543–1.166	0.241	42.1 %	36.26	< 0.001	77.94	0.137
	Quality score											
	Highest quality	10	5759	6916	1.015	0.894–1.153	0.813	72.7 %	15.35	0.082	41.35	0.669
	Lowest quality	7	1249	1394	0.667	0.360–1.236	0.198	27.3 %	36.08	< 0.001	83.37	0.167

Abbreviations: BRCA: Breast carcinoma; OR: odds ratio; CI: confidence interval. Bold values indicate *p*-value < 0.05.

Ultimately, few limitations were raised during the interpretation of our results involving the relatively small sample sizes for the *CYP1A2* (*rs762551*; *c.-9-154 C>A*) variant among the case-control and meta-analysis designs. Additionally, the presence of geographical discrepancies could increase the publication bias and heterogeneity between

studies.

5. Conclusion

In conclusion, this case-control study confirmed the involvement of

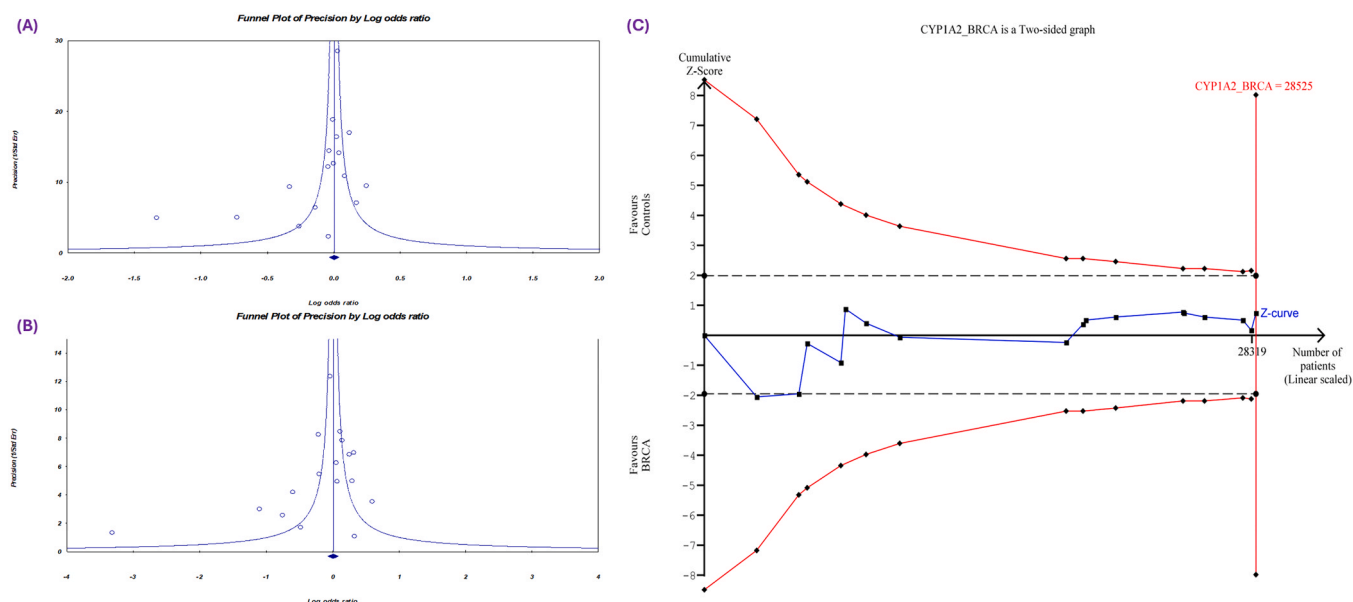


Fig. 5. Publication bias and trial sequential analysis of the studied articles in this meta-analysis design. (A): Funnel plot of precision by Log odds ratio for the association of the *CYP1A2**(rs762551; c.-9-154 C>A) variant with the susceptibility of breast carcinoma under allelic model; (B): Funnel plot of precision by Log odds ratio for the association of the *CYP1A2**(rs762551; c.-9-154 C>A) variant with the susceptibility of breast carcinoma under dominant model; and (C): Trial sequential analysis for the *CYP1A2**(rs762551; c.-9-154 C>A) variant and BRCA susceptibility under the dominant model.

the *CYP1A2**rs762551 variant with a decreased risk of breast carcinoma among Egyptian women. The *CYP1A2**rs762551 variant conferred protection against BRCA development under allelic, dominant, and recessive models. In terms of the meta-analysis design, this study did not achieve significant differences between the *CYP1A2* (rs762551; c.-9-154 C>A) variant and the elevated risk of breast carcinoma among all overall populations. Interestingly, this intronic variant revealed a significant association with a decreased risk of BRCA progression among Caucasian women. Lastly, further records with larger sample sizes and multi-ethnic are required for more precise findings for the *CYP1A2* (rs762551; c.-9-154 C>A) variant with cancer development.

Ethical approval

This work was conducted in accordance with the ethical protocols of the scientific committee and the 1964 Helsinki announcement and its arrangements.

CRediT authorship contribution statement

Maeidh A. Alotaibi: Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Fahad Mohammed Alzuaibr:** Writing – original draft, Validation, Methodology, Formal analysis. **Mohamed Sakran:** Validation, Supervision, Software, Methodology, Data curation. **Adel I. Alalawy:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Rami Elshazli:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no conflict of interest regarding this research.

Acknowledgement

The authors extend their appreciation to the Deanship of Scientific Research at University of Tabuk for funding this work through research

no. S-1442-0102.

Author agreement statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other people who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process.

He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs

Informed consent

Informed consent was obtained from all female subjects registered in this work.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.prp.2024.155501](https://doi.org/10.1016/j.prp.2024.155501).

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